

$$H_0$$

$$\beta_i = 0$$

vs

H_a

$$\underline{\underline{\beta_i \neq 0}}$$

β_i significant.

Model selection

$X_1 \quad X_2 \quad X_3$

$$\left\{ \begin{array}{l} Y = \beta_0 \\ Y_1 = \beta_0 + \beta_1 X_1 \\ Y_2 = \beta_0 + \beta_1 X_1 + \beta_2 X_2 \\ Y_3 = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 \end{array} \right.$$

Setup

Consider the problem of choosing a suitable model given data

$$(y_1, \mathbf{x}_1), (y_2, \mathbf{x}_2), \dots, (y_n, \mathbf{x}_n)$$

where $\mathbf{x}$ represents the vector of predictor variables.

$$\underline{x_1 \dots x_n}$$

Selecting predictors

In the process of selecting the smallest 'best-fitting' model, we are confronted with two contradictory criteria:

- Exclusion of important terms clearly leads to an incorrect model, which can lead to misleading conclusions
(under specified model)
- Inclusion of unnecessary terms diminishes the value of the model as a simplification of the data, and also reduces the statistical accuracy of parameter estimates and predictions.
(over specified model)

Model selection

How do we decide on the best model for our data?

Two parts:

- Choice of procedure
- Choice of criteria

$k \uparrow$ X
 $2^k \uparrow$ X

{ forward selection
backward selection
step wise selection

$$k \uparrow X_i \quad i=1 \dots k$$

The forward selection algorithm with P-values

1. Begin with the null model. $Y = \beta_0$

2. For every term not currently included in the model, calculate a P-value for the inclusion of that term.

3. If the smallest P-value is less than the threshold p_{in} (usually chosen to be 0.05), add that term to the model.

4. Iterate (2), (3) until no further terms are significant.

$$Y = \beta_0 + \beta_2 X_2$$

k models

$$\begin{cases} Y_1 = \beta_0 + \beta_1 X_1 \\ \vdots \\ Y_k = \beta_0 + \beta_k X_k \end{cases}$$

$$F \rightarrow p\text{-value}$$

Example 4.1

The marks data contains the assignment and quiz scores (in percentage) of 339 students in a Statistics course.

Suppose we are interested in the following variables:

- E (response)
- OQ
- A1
- A2
- A3
- A4
- A5
- A6

Fit a multiple linear regression to the data using forward selection.

Example 4.1 Solution

```
marks <- read.csv("marks.csv")
null <- lm(E ~ 1, data=marks)
scope <- E ~ OQ + A1 + A2 + A3 + A4 + A5 + A6
add1(null, scope = scope, test = "F")
```

```
## Single term additions
##
## Model:
## E ~ 1
##
```

	Df	Sum of Sq	RSS	AIC	F value	Pr(>F)
<none>			21.155	-938.43		
OQ	1	7.1258	14.029	-1075.67	171.175	< 2.2e-16 ***
A1	1	1.0852	20.070	-954.28	18.223	2.558e-05 ***
A2	1	1.7407	19.414	-965.54	30.215	7.644e-08 ***
A3	1	4.2472	16.908	-1012.40	84.654	< 2.2e-16 ***
A4	1	7.1621	13.993	-1076.55	172.492	< 2.2e-16 ***
A5	1	6.9001	14.255	-1070.26	163.129	< 2.2e-16 ***
A6	1	9.3016	11.853	-1132.80	264.456	< 2.2e-16 ***

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

$H_0: \gamma = \beta_0$

$H_a: \gamma_1 = \beta_0 + \beta_1 OQ$
 $\gamma_2 = \beta_0 + \beta_1 A_1$

Example 4.1 Solution

```
fs1 <- update(null, .~. + A6)
add1(fs1, scope = scope, test = "F")
```

$$Y = \beta_0 + \beta_1 A_6$$

VS

$$Y = \beta_0 + \beta_1 A_6 + \beta_2 \begin{cases} A_1 \\ \vdots \\ A_5 \end{cases}$$

```
## Single term additions
##
## Model:
## E ~ A6
##
##      Df Sum of Sq    RSS      AIC F value    Pr(>F)
## <none>                11.853 -1132.8
## OQ      1   1.12811  10.725 -1164.7  35.3421 6.929e-09 ***
## A1      1   0.08001  11.773 -1133.1   2.2834  0.13170
## A2      1   0.09538  11.758 -1133.5   2.7255  0.09969 .
## A3      1   0.56043  11.293 -1147.2  16.6749 5.550e-05 ***
## A4      1   0.55393  11.299 -1147.0  16.4720 6.146e-05 ***
## A5      1   0.32088  11.532 -1140.1   9.3489  0.00241 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Example 4.1 Solution

$$Y = \beta_0 + \beta_1 A6 + \beta_2 OQ$$

```
fs2 <- update(fs1, .~. + OQ)
add1(fs2, scope = scope, test = "F")
```

```
## Single term additions
##
## Model:
## E ~ A6 + OQ
##           Df Sum of Sq    RSS      AIC F value    Pr(>F)
## <none>                10.725 -1164.7
## A1           1   0.03104 10.694 -1163.7   0.9725 0.324774
## A2           1   0.02419 10.701 -1163.5   0.7573 0.384812
## A3           1   0.33372 10.391 -1173.4  10.7586 0.001147 **
## A4           1   0.18839 10.537 -1168.7   5.9895 0.014904 *
## A5           1   0.09645 10.629 -1165.8   3.0401 0.082150 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Example 4.1 Solution

$$H_0: Y = \beta_0 + \beta_1 A6 + \beta_2 OQ + \beta_3 A3$$

```
fs3 <- update(fs2, .~. + A3)
add1(fs3, scope = scope, test = "F")
```

V_5

```
## Single term additions
##
## Model:
## E ~ A6 + OQ + A3
##      Df Sum of Sq    RSS      AIC F value Pr(>F)
## <none>                10.391 -1173.4
## A1      1  0.000108  10.391 -1171.4   0.0035 0.9530
## A2      1  0.006228  10.385 -1171.6   0.2003 0.6548
## A4      1  0.070882  10.320 -1173.8   2.2939 0.1308
## A5      1  0.039884  10.351 -1172.7   1.2869 0.2574
```

$$H_a: Y = \beta_0 + \beta_1 A6 + \beta_2 OQ + \beta_3 A3 + \beta_4 \begin{cases} A1 \\ A2 \\ A4 \\ A5 \end{cases}$$

Example 4.1 Solution

Summary(fs3)

```
##
## Call:
## lm(formula = E ~ A6 + OQ + A3, data = stats_marks)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.81856 -0.06018  0.02859  0.09063  0.60694
##
## Coefficients:
##      (Intercept)      Estimate Std. Error t value Pr(>|t|)
## A6              0.13219      0.03273    4.039 6.65e-05 ***
## OQ              0.36301      0.04104    8.845 < 2e-16 ***
## A3              0.20085      0.03726    5.391 1.33e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1761 on 335 degrees of freedom
## Multiple R-squared:  0.5088, Adjusted R-squared:  0.5044
## F-statistic: 115.7 on 3 and 335 DF.  p-value: < 2.2e-16
```

$$y = 0.1321 + 0.36A6 + 0.20OQ + 0.14A3$$

β_0
 β_1
 β_2
 β_3

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.13219	0.03273	4.039	6.65e-05 ***
A6	0.36301	0.04104	8.845	< 2e-16 ***
OQ	0.20085	0.03726	5.391	1.33e-07 ***
A3	0.14387	0.04386	3.280	0.00115 **

$\beta_0 = 0$ vs $\beta_0 \neq 0$
 $\beta_1 = 0$ vs $\beta_1 \neq 0$
 $\beta_2 = 0$ vs $\beta_2 \neq 0$
 $\beta_3 = 0$ vs $\beta_3 \neq 0$

small

$$T = \frac{\beta_0 - \beta_{00}}{SE(\beta_0)}$$

$$H_0: Y = \beta_0 \quad H_a: Y = \beta_0 + \beta_1 A6 + \beta_2 OQ + \beta_3 A3$$

$$\beta_1 = \beta_2 = \beta_3 = 0$$

The backward selection algorithm with P-values

1. Begin with the most complicated model to be considered.
2. For every term currently included in the model, calculate a P-value for the removal of that term.
3. If the largest P-value is greater than the threshold p_{out} (usually chosen to be 0.05), remove that term to the model.
4. Iterate (2), (3) until the model contains only significant terms.

Example 4.2

Consider again the marks data in Example 4.1.

Now fit a multiple linear regression to the data using backward selection.

Example 4.2 Solution

```
full <- lm(E ~ OQ + A1 + A2 + A3 + A4 + A5 + A6
, data=marks)
drop1(full, test = "F")
```

```
## Single term deletions
##
## Model:
## E ~ OQ + A1 + A2 + A3 + A4 + A5 + A6
##      Df Sum of Sq    RSS      AIC F value    Pr(>F)
## <none>          10.280 -1169.1
## OQ      1    0.64466 10.925 -1150.4 20.7558 7.346e-06 ***
## A1      1    0.00357 10.284 -1170.9  0.1149  0.73487
## A2      1    0.01594 10.296 -1170.5  0.5132  0.47427
## A3      1    0.18896 10.470 -1164.9  6.0838  0.01415 *
## A4      1    0.06291 10.343 -1169.0  2.0255  0.15562
## A5      1    0.02283 10.303 -1170.3  0.7352  0.39182
## A6      1    1.06880 11.349 -1137.5 34.4120 1.081e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

drop1(min F, max P-value!)

H_0 74 reduced model H_a Full model

Example 4.2 Solution

```
bs1 <- update(full, .~. -A1)
drop1(bs1, test = "F")
```

```
## Single term deletions
##
## Model:
## E ~ OQ + A2 + A3 + A4 + A5 + A6
##           Df Sum of Sq    RSS      AIC F value    Pr(>F)
## <none>                 10.284 -1170.9
## OQ      1    0.65095 10.935 -1152.1  21.0145 6.464e-06 ***
## A2      1    0.01280 10.297 -1172.5   0.4133 0.52075
## A3      1    0.20414 10.488 -1166.3   6.5901 0.01069 *
## A4      1    0.06108 10.345 -1170.9   1.9720 0.16117
## A5      1    0.02263 10.307 -1172.2   0.7306 0.39330
## A6      1    1.08120 11.365 -1139.0 34.9042 8.579e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Example 4.2 Solution

```
bs2 <- update(bs1, .~. -A2)
drop1(bs2, test = "F")
```

```
## Single term deletions
##
## Model:
## E ~ OQ + A3 + A4 + A5 + A6
##      Df Sum of Sq  RSS      AIC F value    Pr(>F)
## <none>            10.297 -1172.5
## OQ      1   0.64366 10.941 -1154.0  20.8160 7.118e-06 ***
## A3      1   0.19343 10.490 -1168.2   6.2554 0.01286 *
## A4      1   0.05453 10.351 -1172.7   1.7634 0.18511
## A5      1   0.02353 10.320 -1173.8   0.7609 0.38366
## A6      1   1.07782 11.375 -1140.8  34.8565 8.750e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Example 4.2 Solution

```
bs3 <- update(bs2, .~. -A5)
drop1(bs3, test = "F")
```

```
## Single term deletions
##
## Model:
## E ~ OQ + A3 + A4 + A6
##
```

	Df	Sum of Sq	RSS	AIC	F value	Pr(>F)	
## <none>			10.320	-1173.8			
## OQ	1	0.70510	11.025	-1153.3	22.8190	2.670e-06	***
## A3	1	0.21622	10.537	-1168.7	6.9974	0.008549	**
## A4	1	0.07088	10.391	-1173.4	2.2939	0.130826	
## A6	1	1.53773	11.858	-1128.7	49.7654	1.004e-11	***

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Example 4.2 Solution

```
bs4 <- update(bs3, .~. -A4)
drop1(bs4, test = "F")
```

```
## Single term deletions
##
## Model:
## E ~ OQ + A3 + A6
##      Df Sum of Sq    RSS      AIC F value    Pr(>F)
## <none>            10.391 -1173.4
## OQ      1    0.90140 11.293 -1147.2   29.060 1.327e-07 ***
## A3      1    0.33372 10.725 -1164.7   10.759 0.001147 **
## A6      1    2.42664 12.818 -1104.3   78.231 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Example 4.2 Solution

Summary (bs4)

lm

```
##  
## Call:  
## lm(formula = E ~ OQ + A3 + A6, data = stats_marks)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -0.81856 -0.06018  0.02859  0.09063  0.60694   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept)  0.13219    0.03273   4.039 6.65e-05 ***  
## OQ           0.20085    0.03726   5.391 1.33e-07 ***  
## A3           0.14387    0.04386   3.280 0.00115 **   
## A6           0.36301    0.04104   8.845 < 2e-16 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 0.1761 on 335 degrees of freedom  
## Multiple R-squared:  0.5088, Adjusted R-squared:  0.5044   
## F-statistic: 115.7 on 3 and 335 DF,  p-value: < 2.2e-16
```

The stepwise selection algorithm with P-values

1. Begin with the null model. $\gamma = \beta_0$
2. Perform one step of forward selection using a liberal value of p_{in} such as 0.2 or 0.15. 0.1
3. Perform one step of backward elimination with a value of P_{out} such as 0.05.
4. Iterate (2), (3) until no further changes occur or the algorithm cycles.

Example 4.3

Consider again the marks data in Example 4.1.

Fit a multiple linear regression to the data using stepwise selection.

Example 4.3 Solution

2 in 0.10
2 out 0.05

```
add1(null, scope = scope, test = "F")
```

```
## Single term additions
##
## Model:
## E ~ 1
##           Df Sum of Sq    RSS      AIC F value    Pr(>F)
## <none>                21.155   -938.43
## OQ          1      7.1258 14.029 -1075.67 171.175 < 2.2e-16 ***
## A1          1      1.0852 20.070  -954.28  18.223 2.558e-05 ***
## A2          1      1.7407 19.414  -965.54  30.215 7.644e-08 ***
## A3          1      4.2472 16.908 -1012.40  84.654 < 2.2e-16 ***
## A4          1      7.1621 13.993 -1076.55 172.492 < 2.2e-16 ***
## A5          1      6.9001 14.255 -1070.26 163.129 < 2.2e-16 ***
## A6          1      9.3016 11.853 -1132.80 264.456 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Example 4.3 Solution

```
ss1 <- update(null, .~. +A6)
drop1(ss1, test = "F")
```

```
## Single term deletions
##
## Model:
## E ~ A6
##
```

	Df	Sum of Sq	RSS	AIC	F value	Pr(>F)
<none>			11.853	-1132.80		
A6	1	9.3016	21.155	-938.43	264.46	< 2.2e-16 ***

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Example 4.3 Solution

```
add1(ss1, scope=scope, test = "F")
```

```
## Single term additions
##
## Model:
## E ~ A6
##
```

	Df	Sum of Sq	RSS	AIC	F value	Pr(>F)	
## <none>			11.853	-1132.8			
## OQ	1	1.12811	10.725	-1164.7	35.3421	6.929e-09	***
## A1	1	0.08001	11.773	-1133.1	2.2834	0.13170	
## A2	1	0.09538	11.758	-1133.5	2.7255	0.09969	.
## A3	1	0.56043	11.293	-1147.2	16.6749	5.550e-05	***
## A4	1	0.55393	11.299	-1147.0	16.4720	6.146e-05	***
## A5	1	0.32088	11.532	-1140.1	9.3489	0.00241	**

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Example 4.3 Solution

$$Y_i = \beta_0 + \beta_1 A_{6i} + \beta_2 OQ_i$$

```
ss2 <- update(ss1, .~. + OQ)
drop1(ss2, test = "F")
```

```
## Single term deletions
##
## Model:
## E ~ A6 + OQ
##           Df Sum of Sq    RSS      AIC F value    Pr(>F)
## <none>                10.725 -1164.7
## A6         1     3.3039 14.029 -1075.7 103.506 < 2.2e-16 ***
## OQ         1     1.1281 11.853 -1132.8  35.342 6.929e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Example 4.3 Solution

```
add1(ss2, scope=scope, test = "F")
```

```
## Single term additions
##
## Model:
## E ~ A6 + OQ
##      Df Sum of Sq    RSS      AIC F value    Pr(>F)
## <none>                10.725 -1164.7
## A1      1    0.03104 10.694 -1163.7   0.9725 0.324774
## A2      1    0.02419 10.701 -1163.5   0.7573 0.384812
## A3      1    0.33372 10.391 -1173.4  10.7586 0.001147 **
## A4      1    0.18839 10.537 -1168.7   5.9895 0.014904 *
## A5      1    0.09645 10.629 -1165.8   3.0401 0.082150 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Example 4.3 Solution

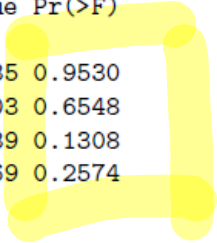
```
ss3 <- update(ss2, .~. + A3)
drop1(ss3, test = "F")
```

```
## Single term deletions
##
## Model:
## E ~ A6 + OQ + A3
##      Df Sum of Sq    RSS      AIC F value    Pr(>F)
## <none>            10.391 -1173.4
## A6      1   2.42664 12.818 -1104.3  78.231 < 2.2e-16 ***
## OQ      1   0.90140 11.293 -1147.2  29.060 1.327e-07 ***
## A3      1   0.33372 10.725 -1164.7  10.759 0.001147 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Example 4.3 Solution

```
add1(ss3, scope=scope, test = "F")
```

```
## Single term additions
##
## Model:
## E ~ A6 + OQ + A3
##      Df Sum of Sq    RSS      AIC F value Pr(>F)
## <none>          10.391 -1173.4
## A1      1  0.000108 10.391 -1171.4  0.0035 0.9530
## A2      1  0.006228 10.385 -1171.6  0.2003 0.6548
## A4      1  0.070882 10.320 -1173.8  2.2939 0.1308
## A5      1  0.039884 10.351 -1172.7  1.2869 0.2574
```



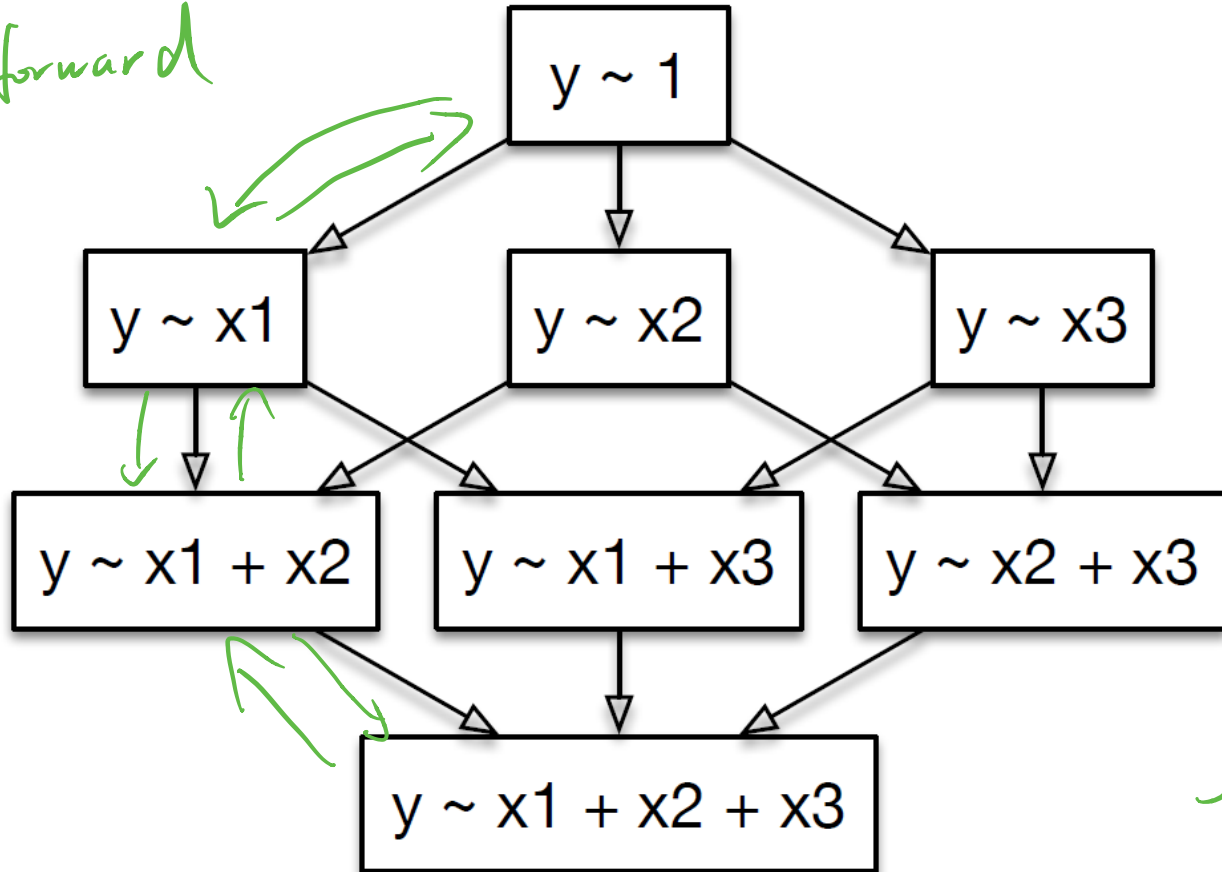
Example 4.3 Solution

```
summary(ss3)
```

```
##
## Call:
## lm(formula = E ~ A6 + OQ + A3, data = stats_marks)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.81856 -0.06018  0.02859  0.09063  0.60694
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.13219    0.03273   4.039 6.65e-05 ***
## A6             0.36301    0.04104   8.845 < 2e-16 ***
## OQ             0.20085    0.03726   5.391 1.33e-07 ***
## A3             0.14387    0.04386   3.280  0.00115 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1761 on 335 degrees of freedom
## Multiple R-squared:  0.5088, Adjusted R-squared:  0.5044
## F-statistic: 115.7 on 3 and 335 DF, p-value: < 2.2e-16
```

Comparison of methods

forward



backward

Principle of marginality

Whenever an interaction term is included in the model, all implied lower order interactions and main effects must also be included.

For example, if we find that we have an interaction term $x_{i1}x_{i2}$ in the model, then we must keep the main effects x_{i1} and x_{i2} in the model.

Some further remarks:

- None of the three methods is guaranteed to produce the “best” model
- They do not necessarily give the same subset of predictors
- In stepwise selection, we can use different p_{in} and p_{out} .
 - If $p_{out} < p_{in}$, we may remove predictors that were added before
 - If $p_{out} > p_{in}$, then this provides some ‘protection’ for the predictors that were added to the model
- Weakness of forward selection
 - Typically begins with an incorrect model, hence significance calculations may be wrong
 - Terms included in the model that may become insignificant later cannot be removed
 - Can fail to detect predictors that are jointly, but not separately, significant

- Weakness of backward selection
 - Assumes the initial model is correct. If this is not correct, we cannot add additional predictors.
 - Can be impractical if there are many predictors
- Stepwise selection combines the good features of forward and backward selection, but still not fool-proof

→ P-value of F -test

Criteria for model comparison

Criteria for comparing models

There are a number of commonly used criteria for comparing models:

- R^2
- Adjusted R^2
- Akaike information criterion (AIC)
- Bayesian information criterion (BIC)

R^2

The proportion of variation in Y (about $\hat{Y}$) explained by the predictors in the model is known as the **coefficient of multiple determination**, denoted by R^2 ,

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}.$$

- larger R^2 is preferred
- Useful for comparing models with same number of predictors

Adjusted R^2

The adjusted R^2 is the R^2 'corrected' for degrees of freedom:

$$\tilde{R}^2 = 1 - \frac{MSE}{MST} = 1 - \left(\frac{n-1}{n-p} \right) (1 - R^2).$$

- larger $\tilde{R}^2$ is preferred
- can compare models of different size

Akaike information criterion (AIC)

The Akaike information criterion (AIC) is defined as

$$AIC = 2p - 2 \ln(\hat{L}),$$

where p is the number of parameter in the model, and $\ln(\hat{L})$ is the log likelihood evaluated at the maximum likelihood estimates.

- the model with the lowest AIC is preferred

$k=2$

Bayesian information criterion (BIC)

A more stringent criterion with respect to the number of parameters is the Bayesian information criterion (BIC). It is defined as

$$BIC = \ln(n)p - 2 \ln(\hat{L}).$$

Handwritten note: A red arrow points from the text "# of obs." to the variable 'n' in the equation.

- the model with the lowest BIC is preferred

For AIC, $k=2$ (R default)

For BIC, $k = \ln(n)$

Example 4.4

Consider again the marks data in Example 4.1.

Fit a multiple linear regression to the data using AIC and

- (a) Forward selection
- (b) Backward selection
- (c) Stepwise selection

Example 4.4 Solutions

```
fAIC <- step(null, scope=scope, direction="forward")
```

back word
"both" ✓

If use BIC. $k = \log(n) \Rightarrow BIC$

```
## Start:  AIC=-938.43
## E ~ 1
##
##      Df Sum of Sq  RSS    AIC
## + A6    1    9.3016 11.853 -1132.80
## + A4    1    7.1621 13.993 -1076.55
## + OQ    1    7.1258 14.029 -1075.67
## + A5    1    6.9001 14.255 -1070.26
## + A3    1    4.2472 16.908 -1012.40
## + A2    1    1.7407 19.414  -965.54
## + A1    1    1.0852 20.070  -954.28
## <none>                21.155  -938.43
##
## Step:  AIC=-1132.8
## E ~ A6
##
##      Df Sum of Sq  RSS    AIC
## + OQ    1    1.1281 10.725 -1164.7
## + A3    1    0.5604 11.293 -1147.2
## + A4    1    0.5539 11.299 -1147.0
## + A5    1    0.3208 11.532 -1140.1
## + A2    1    0.0953 11.758 -1133.5
## + A1    1    0.0800 11.773 -1133.1
## <none>                11.853 -1132.8
```

Example 4.4 Solutions (cont.)

```
##
## Step: AIC=-1164.71
## E ~ A6 + OQ
##
##      Df Sum of Sq  RSS    AIC
## + A3   1  0.33372 10.391 -1173.4
## + A4   1  0.18839 10.537 -1168.7
## + A5   1  0.09645 10.629 -1165.8
## <none>      10.725 -1164.7
## + A1   1  0.03104 10.694 -1163.7
## + A2   1  0.02419 10.701 -1163.5
##
## Step: AIC=-1173.43
## E ~ A6 + OQ + A3
##
##      Df Sum of Sq  RSS    AIC
## + A4   1  0.070882 10.320 -1173.8
## <none>      10.391 -1173.4
## + A5   1  0.039884 10.351 -1172.7
## + A2   1  0.006228 10.385 -1171.6
## + A1   1  0.000108 10.391 -1171.4
```

Example 4.4 Solutions (cont.)

```
##  
## Step:  AIC=-1173.75  
## E ~ A6 + OQ + A3 + A4  
##  
##           Df Sum of Sq    RSS    AIC  
## <none>             10.320 -1173.8  
## + A5      1 0.0235297 10.297 -1172.5  
## + A2      1 0.0136998 10.307 -1172.2  
## + A1      1 0.0003186 10.320 -1171.8
```

Example 4.4 Solutions (cont.)

```
summary(fAIC)
```

```
##
## Call:
## lm(formula = E ~ A6 + OQ + A3 + A4, data = stats_marks)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.81971 -0.06460  0.02923  0.09030  0.61607
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.12306     0.03321   3.705 0.000247 ***
## A6           0.32917     0.04666   7.054 1.00e-11 ***
## OQ           0.18475     0.03868   4.777 2.67e-06 ***
## A3           0.12196     0.04611   2.645 0.008549 **
## A4           0.08014     0.05291   1.515 0.130826
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1758 on 334 degrees of freedom
## Multiple R-squared:  0.5121, Adjusted R-squared:  0.5063
## F-statistic: 87.66 on 4 and 334 DF,  p-value: < 2.2e-16
```

Model validation

Q: Is our model generalizable beyond our sample data?

Ideally, we want to get new observations to validate our model:

(1) Refit the model with the new observations included.

Check to see if the parameter of the new model is significantly different from our previous model.

(2) Use our original model to predict the response of the new observations. Calculate the prediction errors for the new observations.

In practice, it may be difficult to get new observations. An alternative is to split our data into a training set and a validation/testing set. This is called cross-validation (CV).

(1) Fit our model using the training set

(2) Calculate prediction error using the testing set.

Cross-validation

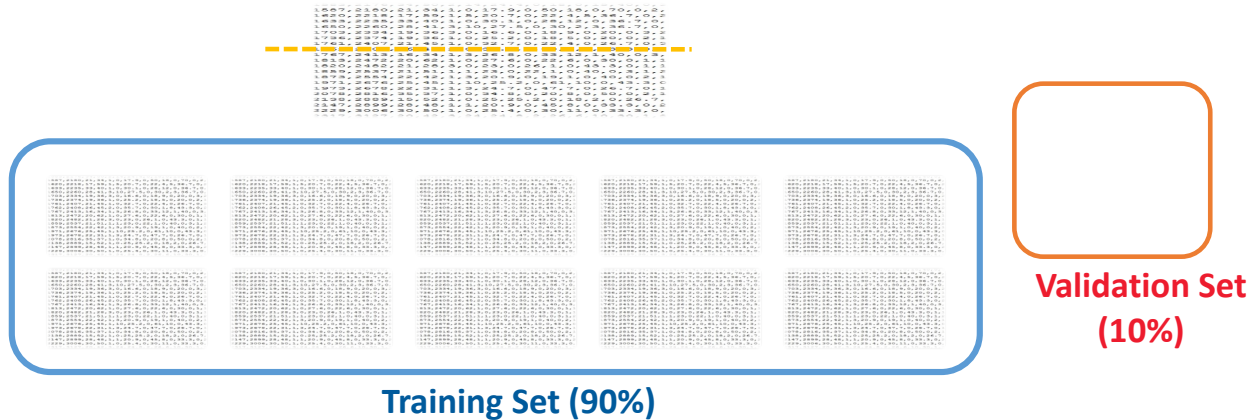
Another useful method to assess how good a model is for prediction is cross-validation.

For k -fold cross-validation, you split the data into k parts.

In each step, train the model on $k - 1$ parts and test for k th part.

Cross-validation

- 10-fold cross-validation



- **Training set:** for model development
- **Validation set:** to assess the accuracy on new data
- **Repeat 10 times:** estimate prediction error

Notation

Label each part

$$C_1, C_2, \dots, C_K$$

Let the number of observations in C_k be n_k ($k = 1, 2, \dots, K$) so that

$$n = \sum_{k=1}^K n_k ,$$

i.e. n is the total number of observations.

Prediction error

The cross validation estimate of the prediction error is

$$CV_{(k)} = \sum_{k=1}^K \frac{n_k}{n} MSE_k,$$

where

$$MSE_k = \sum_{i \in C_k} \frac{(y_i - \hat{y}_i)^2}{n_k},$$

where $\hat{y}_i$ is the fitted value for observation i for the model with part k removed.